# Difficult Airway Management MCQ – Chapter 5

**Source:** Short Textbook of Anesthesia – Ajay Yadav

Section 3: Quintessential in Anesthesia (Chapter 5)

**50 Questions • 30 Minutes**

**Instructions:**
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

# Questions

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| **Q1. What is the definition of a difficult airway?**<br>**A.** Any airway requiring general anesthesia<br>**B.** A clinical situation where a trained anesthesiologist experiences difficulty with face mask ventilation, intubation, or both<br>**C.** An airway requiring fiber-optic intubation<br>**D.** Any airway in a pediatric patient | **Ans: B** |
| **Q2. Which of the following is a congenital cause of difficult intubation?**<br>**A.** Rheumatoid arthritis<br>**B.** Pierre Robin syndrome<br>**C.** Acromegaly<br>**D.** Ludwig's angina | **Ans: B** |
| **Q3. Treacher Collins syndrome causes difficult intubation due to:**<br>**A.** Large tongue<br>**B.** Mandibulofacial dysostosis with micrognathia and malar hypoplasia<br>**C.** Cervical spine fusion<br>**D.** Tracheal stenosis | **Ans: B** |
| **Q4. Which acquired condition can cause difficult airway due to limited mouth opening?**<br>**A.** Marfan syndrome<br>**B.** Temporomandibular joint ankylosis<br>**C.** Down syndrome<br>**D.** Turner syndrome | **Ans: B** |
| **Q5. Acromegaly causes difficult airway due to:**<br>**A.** Short neck<br>**B.** Macroglossia and subglottic narrowing<br>**C.** Cervical spine instability<br>**D.** Tracheomalacia | **Ans: B** |
| **Q6. Which of the following infections can cause acute difficult airway?**<br>**A.** Pneumonia<br>**B.** Ludwig's angina<br>**C.** Urinary tract infection<br>**D.** Cellulitis of the leg | **Ans: B** |
| **Q7. Difficult intubation is defined as requiring more than how many attempts at laryngoscopy?**<br>**A.** 1 attempt<br>**B.** 2 attempts<br>**C.** 3 attempts<br>**D.** 5 attempts | **Ans: C** |

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| **Q8.** **The Mallampati classification is performed with the patient in which position?**<br>**A.** Supine with neck extended<br>**B.** Sitting upright with mouth fully open and tongue protruded without phonation<br>**C.** Left lateral position<br>**D.** Trendelenburg position | **Ans: B** |

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| **Q9.** **In Mallampati Class III, which structures are visible?**<br>**A.** Soft palate, fauces, uvula, and pillars<br>**B.** Soft palate, fauces, and uvula<br>**C.** Soft palate and base of uvula only<br>**D.** Hard palate only | **Ans: C** |

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| **Q10.** **What is the normal thyromental distance in adults?**<br>**A.** Less than 4 cm<br>**B.** Greater than 6.5 cm (3 finger breadths)<br>**C.** 2 cm<br>**D.** 10 cm | **Ans: B** |

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| **Q11.** **The sternomental distance predicting difficult intubation is less than:**<br>**A.** 10 cm<br>**B.** 12.5 cm<br>**C.** 15 cm<br>**D.** 20 cm | **Ans: B** |

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| **Q12.** **Wilson's risk score for difficult intubation includes all of the following EXCEPT:**<br>**A.** Weight<br>**B.** Head and neck movement<br>**C.** Jaw movement<br>**D.** Serum albumin level | **Ans: D** |

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| **Q13.** **The Cormack-Lehane grading system classifies the view at:**<br>**A.** Nasal endoscopy<br>**B.** Direct laryngoscopy<br>**C.** CT scan<br>**D.** Lateral X-ray of neck | **Ans: B** |

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| **Q14.** **Cormack-Lehane Grade I indicates visualization of:**<br>**A.** Only the epiglottis<br>**B.** Entire glottic opening<br>**C.** Posterior commissure only<br>**D.** Neither glottis nor epiglottis | **Ans: B** |

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| **Q15.** **Cormack-Lehane Grade III indicates visualization of:**<br>**A.** Full glottis<br>**B.** Posterior part of glottis<br>**C.** Only the epiglottis<br>**D.** Entire vocal cords | **Ans: C** |

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| **Q16. Cormack-Lehane Grade IV indicates:**<br>**A.** Full view of vocal cords<br>**B.** Partial view of glottis<br>**C.** Only epiglottis visible<br>**D.** Neither glottis nor epiglottis visible | **Ans: D** |

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| **Q17. An inter-incisor distance (mouth opening) of less than how many centimeters suggests difficult intubation?**<br>**A.** 1 cm<br>**B.** 2 cm<br>**C.** 3 cm (2 finger breadths)<br>**D.** 5 cm | **Ans: C** |

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| **Q18. The 'sniffing position' for intubation involves:**<br>**A.** Neck flexion only<br>**B.** Flexion of the lower cervical spine and extension at the atlanto-occipital joint<br>**C.** Full neck extension<br>**D.** Lateral neck rotation | **Ans: B** |

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| **Q19. The BURP maneuver stands for:**<br>**A.** Backward, Upward, Rightward Pressure<br>**B.** Backward, Upward, Rightward Push<br>**C.** Backward, Upward, Rightward Pressure on the larynx<br>**D.** Bilateral, Upward, Rotating Pressure | **Ans: C** |

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| **Q20. Which is the gold standard technique for managing anticipated difficult airway?**<br>**A.** Blind nasal intubation<br>**B.** Awake fiber-optic intubation<br>**C.** Retrograde intubation<br>**D.** Surgical tracheostomy | **Ans: B** |

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| **Q21. The ASA Difficult Airway Algorithm first step in an anticipated difficult airway is:**<br>**A.** Immediate tracheostomy<br>**B.** Awake intubation<br>**C.** Rapid sequence induction<br>**D.** LMA insertion | **Ans: B** |

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| **Q22. Which supraglottic airway device is used as a rescue device in the 'cannot intubate, cannot ventilate' scenario?**<br>**A.** Nasopharyngeal airway<br>**B.** Laryngeal Mask Airway (LMA)<br>**C.** Oropharyngeal airway<br>**D.** Nasal cannula | **Ans: B** |

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| **Q23. In a 'cannot intubate, cannot ventilate' (CICV) emergency, the immediate surgical airway of choice is:**<br>**A.** Tracheostomy<br>**B.** Cricothyrotomy (cricothyroid membrane puncture)<br>**C.** Jet ventilation through nasal cannula<br>**D.** Fiber-optic intubation | **Ans: B** |

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| **Q24. The cricothyroid membrane is located between:**<br>**A.** Thyroid and cricoid cartilages<br>**B.** Cricoid cartilage and first tracheal ring<br>**C.** Hyoid bone and thyroid cartilage<br>**D.** Two tracheal rings | **Ans: A** |

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| **Q25. Which bougie is used as an adjunct during difficult intubation?**<br>**A.** Foley catheter<br>**B.** Gum elastic bougie (Eschmann introducer)<br>**C.** Chest drain<br>**D.** Central venous catheter | **Ans: B** |

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| **Q26. Obesity contributes to difficult airway management due to all EXCEPT:**<br>**A.** Reduced functional residual capacity<br>**B.** Excessive soft tissue in the airway<br>**C.** Difficult positioning<br>**D.** Increased thyromental distance | **Ans: D** |

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| **Q27. Which congenital syndrome is associated with macroglossia causing difficult airway?**<br>**A.** Turner syndrome<br>**B.** Down syndrome (Trisomy 21)<br>**C.** Klinefelter syndrome<br>**D.** Marfan syndrome | **Ans: B** |

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| **Q28. Ankylosing spondylitis causes difficult intubation primarily due to:**<br>**A.** Macroglossia<br>**B.** Fixed flexion deformity of the cervical spine with reduced neck mobility<br>**C.** Trismus<br>**D.** Laryngeal edema | **Ans: B** |

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| **Q29. The 'upper lip bite test' assesses:**<br>**A.** Tongue size<br>**B.** Jaw protrusion and mandibular mobility<br>**C.** Neck extension<br>**D.** Nasal patency | **Ans: B** |

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| **Q30. A short, thick neck is associated with difficult intubation because it:**<br>**A.** Increases the thyromental distance<br>**B.** Limits neck extension and makes alignment of airway axes difficult<br>**C.** Causes vocal cord paralysis<br>**D.** Increases mouth opening | **Ans: B** |

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| **Q31. Which type of laryngoscope blade is preferred for difficult intubation?**<br>**A.** Miller (straight) blade<br>**B.** McCoy (levering tip) blade<br>**C.** Standard Macintosh blade only<br>**D.** No blade can help | **Ans: B** |

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| **Q32. Retrograde intubation involves passing a guide wire through the:**<br>**A.** Nose into the trachea<br>**B.** Cricothyroid membrane upward and out through the mouth<br>**C.** Mouth directly into the trachea<br>**D.** Tracheostomy site | **Ans: B** |

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| **Q33. Which of the following is NOT a predictor of difficult mask ventilation?**<br>**A.** Beard<br>**B.** BMI > 26<br>**C.** Young age<br>**D.** Edentulous patient | **Ans: C** |

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| **Q34. The mnemonic LEMON for airway assessment stands for:**<br>**A.** Look, Evaluate, Mallampati, Obstruction, Neck mobility<br>**B.** Laryngoscopy, Endotracheal, Mask, Oxygen, Nasopharyngeal<br>**C.** Length, Extension, Mouth, Obstruction, Narrowing<br>**D.** Look, Epiglottis, Mandible, Oropharynx, Neck | **Ans: A** |

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| **Q35. The 3-3-2 rule evaluates:**<br>**A.** Number of intubation attempts<br>**B.** Mouth opening (3 fingers), hyoid-mental distance (3 fingers), and thyroid-to-floor of mouth (2 fingers)<br>**C.** Three types of laryngoscope blades<br>**D.** Three airway axes | **Ans: B** |

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| **Q36. Video laryngoscopy is useful in difficult airway because it:**<br>**A.** Eliminates the need for neck extension<br>**B.** Provides an indirect view of the glottis without alignment of airway axes<br>**C.** Can only be used in awake patients<br>**D.** Replaces the need for any backup plan | **Ans: B** |

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| **Q37. For awake fiber-optic intubation, topical anesthesia of the airway is achieved using:**<br>**A.** Bupivacaine 0.5%<br>**B.** Lidocaine 4% (spray/nebulization) and nerve blocks<br>**C.** Thiopentone gargle<br>**D.** Propofol spray | **Ans: B** |
| **Q38. The superior laryngeal nerve block anesthetizes the airway:**<br>**A.** Below the vocal cords<br>**B.** From the epiglottis to the vocal cords (supraglottic area)<br>**C.** Only the nasal cavity<br>**D.** Only the tongue | **Ans: B** |
| **Q39. Transtracheal injection (recurrent laryngeal nerve block) anesthetizes the area:**<br>**A.** Above the vocal cords<br>**B.** Below the vocal cords (infraglottic area)<br>**C.** Nasal cavity<br>**D.** Oral cavity | **Ans: B** |
| **Q40. The intubating LMA (Fastrach) is designed to:**<br>**A.** Replace surgical airway<br>**B.** Facilitate blind or guided tracheal intubation through the LMA<br>**C.** Provide long-term ventilation<br>**D.** Measure tidal volume | **Ans: B** |
| **Q41. The maximum number of intubation attempts recommended before declaring failed intubation is:**<br>**A.** 2<br>**B.** 3<br>**C.** 5<br>**D.** Unlimited | **Ans: B** |
| **Q42. Which of the following tumors can cause difficult airway?**<br>**A.** Renal cell carcinoma<br>**B.** Laryngeal papilloma or carcinoma<br>**C.** Hepatocellular carcinoma<br>**D.** Prostatic carcinoma | **Ans: B** |
| **Q43. Post-radiation therapy to the neck can cause difficult intubation due to:**<br>**A.** Increased mouth opening<br>**B.** Tissue fibrosis, edema, and reduced neck mobility<br>**C.** Enlarged thyroid<br>**D.** Vocal cord hypermobility | **Ans: B** |

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| **Q44. The difficult airway cart should contain all of the following EXCEPT:**<br>**A.** Various sizes of ETT and LMA<br>**B.** Fiber-optic bronchoscope<br>**C.** Cricothyrotomy set<br>**D.** Cardiopulmonary bypass machine | **Ans: D** |

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| **Q45. Preoxygenation before intubation attempts is important because it:**<br>**A.** Makes intubation easier<br>**B.** Provides an oxygen reserve (extends safe apnea time)<br>**C.** Reduces blood pressure<br>**D.** Causes bronchodilation | **Ans: B** |

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| **Q46. In a patient with cervical spine injury, the preferred method of intubation is:**<br>**A.** Conventional direct laryngoscopy with full neck extension<br>**B.** Manual in-line stabilization with fiber-optic or video laryngoscopy<br>**C.** Blind nasal intubation without precautions<br>**D.** Immediate tracheostomy | **Ans: B** |

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| **Q47. Jet ventilation through a cricothyrotomy cannula uses oxygen at a pressure of:**<br>**A.** 1–2 psi<br>**B.** 50 psi (high-pressure source)<br>**C.** 100 psi<br>**D.** Normal wall oxygen at 4 L/min | **Ans: B** |

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| **Q48. Which of the following is a sign of successful tracheal intubation?**<br>**A.** Gurgling sounds on auscultation<br>**B.** Presence of end-tidal $CO_2$ on capnography<br>**C.** Distension of the abdomen<br>**D.** Absence of chest rise | **Ans: B** |

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| **Q49. The lightwand (lighted stylet) technique works on the principle of:**<br>**A.** Fluoroscopy<br>**B.** Transillumination of the anterior neck when the tip is in the trachea<br>**C.** Ultrasound guidance<br>**D.** Magnetic navigation | **Ans: B** |

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| **Q50. Which of the following is the most important step after a failed intubation attempt?**<br>**A.** Immediately attempt again with the same technique<br>**B.** Ensure adequate oxygenation and ventilation before the next attempt<br>**C.** Proceed directly to surgical airway<br>**D.** Extubate the patient | **Ans: B** |

# Answer Key & Explanations

**Q1. Answer: B — A clinical situation where a trained anesthesiologist experiences difficulty with face mask ventilation, intubation, or both**

A difficult airway is defined as a clinical situation in which a conventionally trained anesthesiologist experiences difficulty with face mask ventilation, tracheal intubation, or both.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q2. Answer: B — Pierre Robin syndrome**

Pierre Robin syndrome (micrognathia, glossoptosis, cleft palate) is a congenital cause of difficult intubation.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q3. Answer: B — Mandibulofacial dysostosis with micrognathia and malar hypoplasia**

Treacher Collins syndrome involves mandibulofacial dysostosis with micrognathia, making laryngoscopy and intubation difficult.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q4. Answer: B — Temporomandibular joint ankylosis**

TMJ ankylosis causes severe restriction of mouth opening, making conventional laryngoscopy impossible.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q5. Answer: B — Macroglossia and subglottic narrowing**

Acromegaly leads to macroglossia, prognathism, and subglottic narrowing, all contributing to difficult airway management.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q6. Answer: B — Ludwig's angina**

Ludwig's angina causes submandibular space infection with floor of mouth swelling, leading to airway compromise.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q7. Answer: C — 3 attempts**

Difficult intubation is defined as requiring more than 3 attempts at laryngoscopy or attempts lasting longer than 10 minutes.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q8. Answer: B — Sitting upright with mouth fully open and tongue protruded without phonation**

Mallampati assessment is done with patient sitting, mouth wide open, tongue maximally protruded, without phonation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q9. Answer: C — Soft palate and base of uvula only**

Mallampati Class III shows only the soft palate and base of uvula, suggesting potentially difficult intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q10. Answer: B — Greater than 6.5 cm (3 finger breadths)**

Normal thyromental distance is greater than 6.5 cm (3 finger breadths). Less than 6 cm suggests difficult intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q11. Answer: B — 12.5 cm**

A sternomental distance less than 12.5 cm with the head fully extended and mouth closed is predictive of difficult intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q12. Answer: D — Serum albumin level**

Wilson's risk score includes weight, head/neck movement, jaw movement, receding mandible, and buck teeth. Serum albumin is not part of it.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q13. Answer: B — Direct laryngoscopy**

Cormack-Lehane grading classifies the laryngoscopic view obtained during direct laryngoscopy.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q14. Answer: B — Entire glottic opening**

Grade I shows the entire glottic opening, indicating easy intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q15. Answer: C — Only the epiglottis**

Grade III shows only the epiglottis with no part of the glottis visible, indicating difficult intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q16. Answer: D — Neither glottis nor epiglottis visible**

Grade IV means neither the glottis nor the epiglottis can be visualized – the most difficult scenario.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q17. Answer: C — 3 cm (2 finger breadths)**

A mouth opening less than 3 cm (2 finger breadths) suggests difficulty with laryngoscopy and intubation.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q18. Answer: B — Flexion of the lower cervical spine and extension at the atlanto-occipital joint**

The sniffing position involves flexion of the lower cervical spine (pillow under head) and extension at the atlanto-occipital joint to align the oral, pharyngeal, and laryngeal axes.

*Topic: Management | Ref: Ch 5, Page 54*

**Q19. Answer: C — Backward, Upward, Rightward Pressure on the larynx**

BURP stands for Backward, Upward, and Rightward Pressure on the thyroid cartilage to improve laryngoscopic view.

*Topic: Management | Ref: Ch 5, Page 54*

**Q20. Answer: B — Awake fiber-optic intubation**

Awake fiber-optic intubation is considered the gold standard for managing an anticipated difficult airway.

*Topic: Management | Ref: Ch 5, Page 54*

**Q21. Answer: B — Awake intubation**

The ASA algorithm recommends awake intubation as the primary approach when a difficult airway is anticipated.

*Topic: Management | Ref: Ch 5, Page 54*

**Q22. Answer: B — Laryngeal Mask Airway (LMA)**
The LMA is a key rescue device in failed intubation scenarios and is part of the difficult airway algorithm.

*Topic: Management | Ref: Ch 5, Page 54*

**Q23. Answer: B — Cricothyrotomy (cricothyroid membrane puncture)**
Emergency cricothyrotomy through the cricothyroid membrane is the life-saving procedure in CICV situations.

*Topic: Management | Ref: Ch 5, Page 54*

**Q24. Answer: A — Thyroid and cricoid cartilages**
The cricothyroid membrane lies between the inferior border of the thyroid cartilage and the superior border of the cricoid cartilage.

*Topic: Management | Ref: Ch 5, Page 54*

**Q25. Answer: B — Gum elastic bougie (Eschmann introducer)**
The gum elastic bougie (Eschmann stylet/introducer) is a commonly used adjunct to facilitate tracheal intubation when the glottic view is poor.

*Topic: Management | Ref: Ch 5, Page 54*

**Q26. Answer: D — Increased thyromental distance**
Obesity does not increase thyromental distance. It causes excessive soft tissue, reduced FRC, and difficult positioning.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q27. Answer: B — Down syndrome (Trisomy 21)**
Down syndrome is associated with macroglossia, short neck, and atlantoaxial instability, all contributing to difficult airway.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q28. Answer: B — Fixed flexion deformity of the cervical spine with reduced neck mobility**
Ankylosing spondylitis causes fusion of cervical vertebrae leading to a fixed, flexed cervical spine with severely limited extension.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q29. Answer: B — Jaw protrusion and mandibular mobility**
The upper lip bite test assesses the ability to protrude the mandible (jaw protrusion), which correlates with ease of laryngoscopy.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q30. Answer: B — Limits neck extension and makes alignment of airway axes difficult**
A short, thick neck limits neck extension and makes alignment of the oral, pharyngeal, and laryngeal axes difficult.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q31. Answer: B — McCoy (levering tip) blade**
The McCoy blade has a hinged tip that can be levered to lift the epiglottis, improving the view in difficult laryngoscopy.

*Topic: Management | Ref: Ch 5, Page 54*

**Q32. Answer: B — Cricothyroid membrane upward and out through the mouth**

In retrograde intubation, a wire is passed through the cricothyroid membrane in a cephalad direction, emerging from the mouth, then used as a guide for the ETT.

*Topic: Management | Ref: Ch 5, Page 54*

**Q33. Answer: C — Young age**

Young age is not a predictor of difficult mask ventilation. Beard, obesity, edentulous status, and snoring are predictors.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q34. Answer: A — Look, Evaluate, Mallampati, Obstruction, Neck mobility**

LEMON stands for Look externally, Evaluate 3-3-2 rule, Mallampati, Obstruction, and Neck mobility.

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q35. Answer: B — Mouth opening (3 fingers), hyoid-mental distance (3 fingers), and thyroid-to-floor of mouth (2 fingers)**

The 3-3-2 rule assesses inter-incisor distance (3 fingers), hyoid-mental distance (3 fingers), and thyroid notch-to-floor of mouth (2 fingers).

*Topic: Assessment of Difficult Intubation | Ref: Ch 5, Page 53*

**Q36. Answer: B — Provides an indirect view of the glottis without alignment of airway axes**

Video laryngoscopy provides an indirect view of the glottis on a screen without requiring alignment of the oral, pharyngeal, and laryngeal axes.

*Topic: Management | Ref: Ch 5, Page 54*

**Q37. Answer: B — Lidocaine 4% (spray/nebulization) and nerve blocks**

Topical lidocaine 4% via spray or nebulization, supplemented by nerve blocks (superior laryngeal, transtracheal), provides airway anesthesia for awake FOI.

*Topic: Management | Ref: Ch 5, Page 54*

**Q38. Answer: B — From the epiglottis to the vocal cords (supraglottic area)**

The internal branch of the superior laryngeal nerve supplies sensation from the epiglottis to and including the vocal cords.

*Topic: Management | Ref: Ch 5, Page 54*

**Q39. Answer: B — Below the vocal cords (infraglottic area)**

Transtracheal injection through the cricothyroid membrane anesthetizes the trachea below the vocal cords (recurrent laryngeal nerve territory).

*Topic: Management | Ref: Ch 5, Page 54*

**Q40. Answer: B — Facilitate blind or guided tracheal intubation through the LMA**

The intubating LMA (ILMA/Fastrach) is specifically designed to allow passage of an endotracheal tube through the device.

*Topic: Management | Ref: Ch 5, Page 54*

**Q41. Answer: B — 3**

After 3 failed attempts at intubation by an experienced laryngoscopist, the situation should be declared as failed intubation and the algorithm followed.

*Topic: Management | Ref: Ch 5, Page 54*

**Q42. Answer: B — Laryngeal papilloma or carcinoma**

Laryngeal tumors can distort airway anatomy and obstruct the glottic opening, making intubation difficult.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q43. Answer: B — Tissue fibrosis, edema, and reduced neck mobility**

Radiation to the neck causes tissue fibrosis, reduced mobility, and edema leading to difficult airway management.

*Topic: Causes of Difficult Intubation/Difficult Airway | Ref: Ch 5, Page 52*

**Q44. Answer: D — Cardiopulmonary bypass machine**

A difficult airway cart contains various airway devices (ETTs, LMAs, bougies, fiber-optic scope, cricothyrotomy set) but not a bypass machine.

*Topic: Management | Ref: Ch 5, Page 54*

**Q45. Answer: B — Provides an oxygen reserve (extends safe apnea time)**

Preoxygenation with 100% O2 denitrogenates the FRC, providing an oxygen reserve that extends safe apnea time during intubation attempts.

*Topic: Management | Ref: Ch 5, Page 54*

**Q46. Answer: B — Manual in-line stabilization with fiber-optic or video laryngoscopy**

In cervical spine injury, intubation is performed with manual in-line stabilization (MILS) using fiber-optic or video laryngoscopy to avoid neck movement.

*Topic: Management | Ref: Ch 5, Page 54*

**Q47. Answer: B — 50 psi (high-pressure source)**

Transtracheal jet ventilation uses a high-pressure oxygen source (typically 50 psi) through a narrow cannula in the cricothyroid membrane.

*Topic: Management | Ref: Ch 5, Page 54*

**Q48. Answer: B — Presence of end-tidal CO2 on capnography**

Capnography (detection of end-tidal CO2) is the gold standard for confirming correct tracheal tube placement.

*Topic: Management | Ref: Ch 5, Page 54*

**Q49. Answer: B — Transillumination of the anterior neck when the tip is in the trachea**

The lightwand uses transillumination – a bright glow seen in the midline of the anterior neck confirms the stylet tip is in the trachea.

*Topic: Management | Ref: Ch 5, Page 54*

**Q50. Answer: B — Ensure adequate oxygenation and ventilation before the next attempt**

After a failed attempt, the priority is to ensure adequate oxygenation and ventilation (bag-mask or LMA) before considering the next attempt or alternative plan.

*Topic: Management | Ref: Ch 5, Page 54*